Global semantic banks (TRAIN only)

TRAIN captions train + restval → POS tagging + lemmatization → CLIP text encoder Frozen *

- Fixed noun bank → Learned semantic aggregation → Pn
- Fixed verb bank → Learned semantic aggregation → Pv

Pn, Pv: 150 × 512

Prepared once:
lemmas and fixed banks.
Aggregation remains trainable.

Visual region extraction

Linear projection 768 → 512
ROI pooling
Patch features 14 × 14 × 768
Boxes B (x1, y1, x2, y2)
CLIP ViT-B/16 Visual encoder *
YOLO Detector *



Input image I

V

Semantic bank–region alignment

Object fusion 1 Self / cross-attn + gate
V1, T1
Object fusion 2 Self / cross-attn + gate
Relation fusion Self / cross-attn + gate
Vo
Vr
All region memories: N × 512

Image-specific prompts and region links

UPT detector * Input image I
Object / relation phrases
CLIP text encoder to / tr *
UPT boxes
Class + IoU ≥ 0.5 matching
YOLO boxes B

Text to, tr → both Slot MLPs

Unmatched slots omitted

ROI links → both Slot MLPs / masks

Object prompt adapter

Slot MLP: [to; Vo[i]; bi]
Q
K, V
Masked cross-attn
Output projection before residual sum
Vo′

Relation prompt adapter

Slot MLP: [tr; Vo[s,o]; bs,bo]
Q
K, V
Masked cross-attn
Output projection before residual sum
Vr′

Both slot MLPs use original Vo.
Mask: linked ROI / pair endpoints only.

Caption generation

Vo′ and Vr′ stay separate

Transformer decoder ×3
Masked self-attention
Parallel object / relation cross-attn
Gated fusion + FFN

Token prefix → Word embedding + fixed positional encoding
Vocabulary projection + softmax
Beam search (width 5)
Five candidate captions
D: likelihood · C: CLIP alignment
O: object consistency
S = 0.3D + 0.4C + 0.3O
Select highest-scoring candidate
Final caption

Reranking inputs: image I + YOLO labels

Semantic fusion mechanism (shared structure; separate parameters)

Soft semantic selection Cosine relevance → T
Self-attention Q = K = V = X
Cross-attention Q = H; K = V = T
Learned sigmoid gate Mix H and cross context
H bypass
Updated regions

Fusion includes residual normalization and FFN; adapters include a zero-initialized output projection.

Training: cross-entropy → SCST + cross-entropy. No reference captions at inference.

* Frozen Trainable